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Q1. Food Delivery Billing System (Multiple Inheritance using Two Interfaces)

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
class Main{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        double foodCost=sc.nextDouble();
        String couponCode=sc.next();

        FoodOrder fo=new FoodOrder(foodCost,couponCode);
        double deliveryCharge=fo.calculateDeliveryCharge();
        double discount=fo.calculateDiscount();
        double finalbill=foodCost+deliveryCharge-discount;
        System.out.println("Food Cost : "+foodCost);
        System.out.println("Delivery Charge : "+deliveryCharge);
        System.out.println("Discount : "+discount);
        System.out.println("Final Bill : "+finalbill);
    }
}
interface DeliveryCharge{
    double calculateDeliveryCharge();
}
interface CouponDiscount{
    double calculateDiscount();
}
class FoodOrder implements DeliveryCharge,CouponDiscount{
    double foodCost;
    String couponCode;
    FoodOrder(double foodCost,String couponCode){
        this.foodCost=foodCost;
        this.couponCode=couponCode;
    }
    public double calculateDeliveryCharge(){
        if (foodCost>=500){
            return 0.0;
        }
        else
            return 50.0;
    }
    public double calculateDiscount(){

        if(couponCode.equals("SAVE10")){
            return foodCost*0.10;
        }
        else{
            return 0.0;
        }
    }
}
```

INPUT

600
SAVE10

OUTPUT

Food Cost : 600.0
Delivery Charge : 0.0
Discount : 60.0
Final Bill : 540.0

Q2. Hotel Billing System (Multiple Inheritance using One Class and One Interface)

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
class Main{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        int RoomNumber=sc.nextInt();
        String GuestName=sc.next();
        double RoomRent=sc.nextDouble();
        HotelBill hb=new HotelBill(RoomNumber,GuestName,RoomRent);
        double ServiceCharge=hb.calculateServiceCharge();
        double finalbill=RoomRent+ServiceCharge;
        System.out.println("Room Number : "+RoomNumber);
        System.out.println("Guest Name : "+GuestName);
        System.out.println("Room Rent : "+RoomRent);
        System.out.println("Service Charge : "+ServiceCharge);
        System.out.println("Final Bill : "+finalbill);
    }
}
class Room{
    int RoomNumber;
    String GuestName;
    double RoomRent;
}
interface ServiceCharge{
    double calculateServiceCharge();
}
class HotelBill extends Room implements ServiceCharge{
    HotelBill(int RoomNumber,String GuestName,double RoomRent){
        this.RoomNumber=RoomNumber;
        this.GuestName=GuestName;
        this.RoomRent=RoomRent;
    }
    public double calculateServiceCharge(){
        if(RoomRent>=5000){
            return RoomRent*0.12;
        }
        else{
            return RoomRent*0.08;
        }
    }
}
```

INPUT

```
205
Ananya
6000.0
```

OUTPUT

```
Room Number : 205
Guest Name : Ananya
Room Rent : 6000.0
Service Charge : 720.0
Final Bill : 6720.0
```